2025 5th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC) | 979-8-3315-5445-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICAIRC68035.2025.11385215

A Hybrid Machine Learning Framework Combining Neural Networks and Shapley Decomposition for Olympic Analytics

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Abstract. **This paper presents an integrated analytical framework for Olympic medal forecasting by combining neural networks with causal inference methods. We develop an optimized BP neural network enhanced by bio-inspired algorithms to predict national medal performances, demonstrating superior accuracy through comprehensive validation. The study further examines breakthrough probabilities for non-medal-winning nations and quantifies sport-specific contributions using game-theoretical approaches. A novel causal analysis evaluates coaching effectiveness across different athletic disciplines.**

Our multi-model system provides strategic insights for sports governance, identifying key performance determinants including host advantages, historical achievement patterns, and optimal event participation strategies. The methodology offers robust predictive capabilities while addressing critical challenges in sports analytics through innovative machine learning applications.

Keywords—BP neural networks, Logistics regressions, Shapley Value Decomposition, Bayesian Causal Forest

I. INTRODUCTION

In recent years, the Olympic Games, as the world's top sports event, the total number of gold medals and medals won not only represents the level of a country's competitive sports, but also becomes an important means for many countries to enhance their national soft power and national image by demon- strating their comprehensive economic and social national power. Accurately predicting the total num- ber of Olympic gold medals and the ranking of each country is of great significance to the formulation of our country's goal of participating in the Olympic Games, and it is also a hot topic that people pay attention to before the Olympic Games.

The Paris Olympics, successfully held in Paris, France from July 26 to August 11, 2024, was a global sporting event that brought together top athletes from all over the world to compete for Olympic medals and display sportsmanship and style. The United States ranked first with 126 total medals, along with China with 40 gold medals, and host France ranked fifth in the medal tally. The global audience paid close attention to the top countries, as well as to the medal counts of other countries.

However, the prediction of the final medals is usually made with the announcement of the athletes of the new Games rather than based on historical Olympic data, and there is still a large gap in the prediction research [1].The study construct is Figure 1.

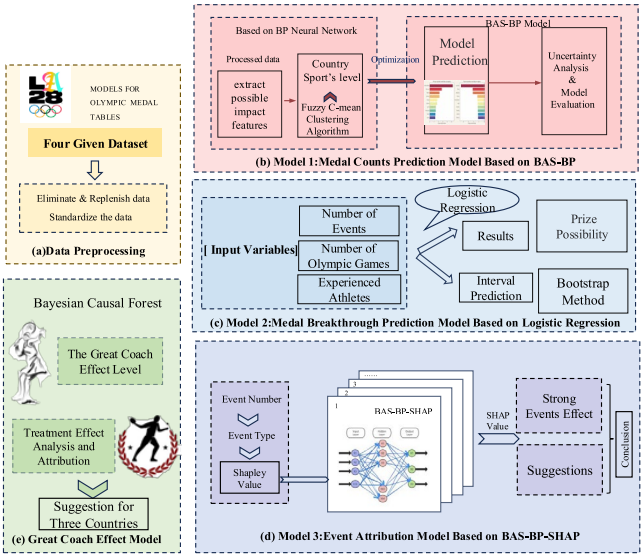


Fig. 1. study design

II. ASSUMPTION AND NOTATION

To simplify the model,we have made several assumptions. However,we may need to relax some if these assumptions to optimize the model and increase its applicability in complex real-world environ- ments.

- we assume that the number of medals won in the first three Olympics reflects the country's level of athleticism in that sport over that period of time.
- Considering the impact of major historical events such as World War I, World War II, the Cold War, and the dissolution of the Soviet Union on the Olympics and its athletes, we assume that using data only after 1991 would improve the accuracy of the model's predictions.
- Due to the lack of direct access to data on GDP, population, etc. for each country, we assume that these factors do not have a significant impact on the final results of our models.
- WeassumethattheperformanceofathletesfromeachcountryisrelativelystableintheOlympic Games they participate in

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The Variables and functions are in the Table I.

TABLE I. VARIABLES AND FUNCTIONS

Variable	definition
Avg_Gold	Average gold medals over the previous three years
Avg_Silver	Average silver medals over the previous three years
Avg_Bronze	Average bronze medals over the previous three years
Gold_this_year	gold medals of this year
Silver_this_year	silver medals of this year
Bronze_this_year	bronze medals of this year
p_host	whether to be the host country
p_post_host	whether to be the host country
p_sport'slevel	whether to be the previous host country
P_1	the number of aquatic events
P_2	the number of track&field events
P_3	the number of ball events
P_4	the number of gymnastics events
P_5	the number of strength events
P_6	the number of cycling events
P_7	the number of shooting events
P_8	the number of other events
Xi	LR model input value
yi	the actual value
$yi*$	the model's predicted output
ϕi	Shapley's value for the feature

III. MEDAL COUNTS PREDICTION MODEL BASED ON BAS-BP

A. Data Preprocessing

- Count gold, silver and bronze medals: count the number of gold, silver and bronze MEDALS won by each country and year.
- home effect variable (host): based on the Host information table (host table), a home effect variable is created for the host country of each Olympic Games.
- Count the number and type of events: Count the number of events in which each country participated in each Olympic Game, and divide them into categories according to aquatic events, track and field events, ball events, gymnastics events, strength events, cycling events, shooting events and other events.
- Normalization Processing:In order to improve the training efficiency, stability and prediction accuracy of the model. It is necessary to normalize the feature variables, that is, scale the data range to [0,1].

$$x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

B. Model Optimization Based on BP

As a classic and basic algorithm in artificial neural networks, Back propagation (BP) neural net- works have been widely used [2]

To accurately predict the medal counts for the 2028 Los Angeles Summer Olympics, we utilize a BP neural network model. However, the random initialization of weights and thresholds in BP networks often causes the training process to converge to local optima, limiting prediction accuracy. To address this, we incorporate the Beetle Antennae Search (BAS) algorithm to optimize the initial weights and thresholds of the BP neural network. All the steps can be founded in the Figure 2.

The Beetle Antennae Search (BAS) algorithm, proposed in 2017, is a meta-heuristic optimization method inspired by the foraging behavior of beetles.

Step 1 Construct a random vector representing the beetle's antenna orientation

Step 2 Define the fitness function: The mean squared error (MSE) is employed as the fitness evaluation function to guaide and advance the research within the solution space.

$$\text{fitness} = \frac{1}{m}\sum_{i=1}^{m}(y_i - y_i^*)^2 \quad (2)$$

Step3 Set the step size factor δ : Unlike classical optimization algorithms such as PSO and GA, the BAS algorithm can accelerate the iteration process by configuring the attenuation coefficient and step size factor [3].

$$\delta^{t+1} = \delta^t \times \text{eta}, t=(0,1,2,3,\ldots,n) \quad (3)$$

Step 4 Search Process: Initialize the beetle's starting position "beatX" randomly within [-0.5, 0.5], and compute its initial fitness "bestY" using Equation (3). Then update the positions of its left and right antennae via Equation (4). Each iteration involves evaluating smell intensity at both antennae and updating the beetle's centroid position, which correspondingly adjusts the weights and thresholds of the BP neural network.

Step 5 Generate the Optimal Solution: The optimal solution is used as the initial weights and thresholds for BP neural network, serving as the starting point for network training

Step 6 Set BP neural network structure: The data sample format consists of 14 inputs and 3 outputs, meaning the neural network contains 14 neurons in the input layer and 3 neurons in the output layer.

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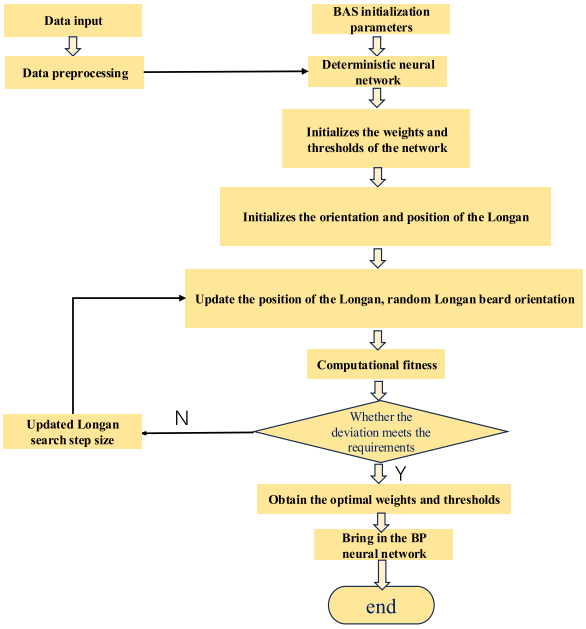


Fig. 2. Forecast Flow Chart of BAS-BP Model

C. Features weight Results of BAS-BP

TABLE II. FEATURE WEIGHT RESULT OF BAS-BP

Feature Weight	*Gold_This_Year*	*Silver_This_Year*	*Bronze_This_Year*
Avg_Gold	0.0011	0.0012	0.0007
Avg_Silver	0.0009	0.0009	0.0013
Avg_Bronze	0.0007	0.0006	0.0007
Aquatics	0.0007	0.0009	0.0012
Track&Field events	0.0009	0.0010	0.0023
Ball events	0.0001	0.0002	0.0007
Gymnastics events	0.0019	0.0021	0.0030
Strength events	-0.0003	0.0003	0.0011
Cycling events	0.0001	0.0013	0.0015
Shooting	0.0027	0.0033	0.0034
Other events	-0.0001	-0.0005	-0.0005
p_host	0.6481	0.6336	0.7027
p_posthost	0.1703	0.1539	0.0534
p_sports_level	0.1756	0.2012	0.2287

The Table II shows the influence of different features on predicting the number of gold, silver and bronze medals won by countries in the 2028 Olympic Games, that is, the weight of each feature in the model. Among them, the host effect (p_host) has the most significant effect on the number of medals, showing the highest weight in the prediction model for gold, silver and bronze medal, which are 0.6481, 0.6336 and 0.7027 respectively. The second is the national sports level (p_sports' level), which gradually increases as the medal level increases.

D. Medal Counts Prediction Model Results&Evaluation

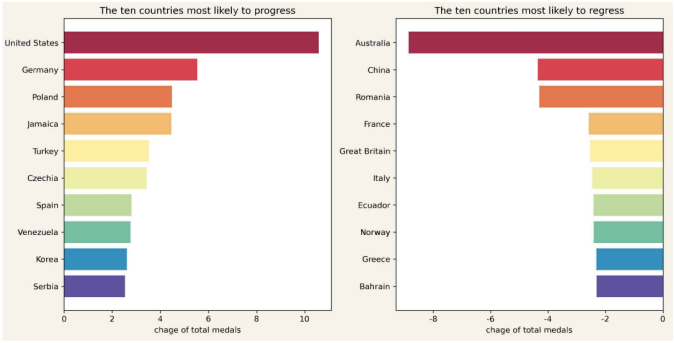


Fig. 3. Prediction Based on BAS-BP

TABLE III. FEATURE WEIGHT RESULT OF BAS-BP

country	*Gold_prediction*	*Total_prediction*	*Confidence interval*
United States	50	137	[36,62]
Germanyr	12	39	[9,15]
Poland	3	14	[2,5]
Jamaica	4	10	[2,5]
Tukey	1	4	[-1,2]
Czechia	1	3	[0,2]
Spain	6	21	[3,9]
Venezuela	1	3	[0,1]
Serbia	3	8	[2,4]

Through the trained model, the number of gold medals and total medals of each country in the 2028 Olympic Games are predicted which can be founded from Figure 3 and Table III.

IV. MEDAL BREAKTHROUGH PREDICTION MODEL BASED ON LO- GISTIC REGRESSION

The logistic regression (LR) model establishes a nonlinear probability function to represent the relationship between independent and dependent variables through appropriate regression coefficients. It is commonly used for statistical analysis of binary dependent variables (where the dependent variable can only be 0 or 1). In this case, whether a country that participated in previous Olympic Games has won a medal is represented as a binary variable: 1 indicates participation with a medal won, and 0 indicates participation without a medal. The prediction of a breakthrough for non-medal-winning countries is essentially a binary classification problem.

Division of Training Set and Test Set

Data for the 2004 to 2024 Olympic Games are used in this article. From the training data, countries that have won the prize

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for the first time are selected as the training set, and countries that have never won the prize are selected as the prediction set.

Logistic Regression Analysis Objectives

The dependent variable is whether the participating country won a medal (logical value 1) or did not win a medal (logical value 0). The independent variables include the total number of events the country participated in ($X1$), the number of Olympic Games the country has participated in ($X2$), and whether this year's participants include athletes with prior Olympic experience ($X3$). Among these, $X3$ is a categorical variable, while $X1$ and $X2$ are continuous variables.

$$\ln\left(\frac{p(Y=1)}{1-p(Y=1)}\right) = \alpha + \sum_{i=1}^{k} \beta_i X_i \quad (2)$$

$$p(Y=1|X) = \frac{\exp\left(\alpha + \sum_{i=1}^{k} \beta_i X_i\right)}{1 + \exp(\alpha + \sum_{i=1}^{k} \beta_i X_i)} \quad (3)$$

(Where Xi represents the i-th variable that affects whether a participating country wins a medal. βi represents the partial regression coefficient of explanatory variable.)

A. Model Training

Use the training dataset to train a Logistic Regression Model and use the Maximum Likelihood Estimation (MLE) method to estimate the model parameters βi.

B. Model Prediction

Prediction sets were extracted from countries that did not win prizes and data were preprocessed. The trained model is used to make predictions on the prediction set to calculate the probability of each country winning the next Olympic Games.the model results is in the Tavle IV.

TABLE IV. MODEL RESULTS

Country	Predicted_probability	Confidence interval
Abgola	0.36	[015,0.69]
Liberia	0.11	[0.05,0.20]
Nicaragua	0.08	[0.03,0.15]
EL Salvador	0.07	[0.03,0.13]
PapaNewGuinea	0.08	[0.03,0.13]

In addition to potential athletic performance, these figures also reflect the likelihood of each country winning a medal at the Olympics. Most of the non-winning countries are located in Africa and Central America, and the probability of winning a breakthrough prize is generally low, as can be seen from the Table IV.

Interval Prediction Based on Bootstrap Method

Logistic regression results show that the probability of these countries winning medals is generally low. Considering the actual situation, the reason why a country has not won a medal is that it is difficult for them to win a medal due to factors such as economic level (GDP) limiting sports input, lack of talents due to small population base, lack of experienced coaches and scientific training programs. Therefore, we set the threshold at a low level to avoid the situation where the predicted value is all 0. Based on multiple bootstrap sampling results (Figure 4) We select a suitable threshold based on the results of multiple predictions, and we get a forecast result of 3 countries (0.2) with a confidence interval of [1,3] for this result, our confidence level is 95%.

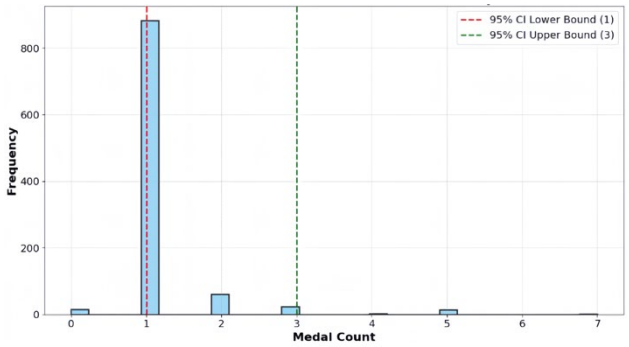


Fig. 4. Dstribution of Medal Counts

V. EVENT ATTRIBUTION MODEL BASED ON BAS-BP-SHAP

The weights of the characteristic indicators of Model 1 based on BAS-BP established according to the first question can reflect the impact of the number of events participating in the participating countries on the number of medals. Based on this, we also need to quantify the categories of events participating in the participating countries and the impact of the host country's new events in this Olympic Games on the number of medals of participating countries and the relationship between them.

Furthermore,we introduce Shapley Value Decomposition to analyze and quantify the impact of event categories on the medal counts of participating countries and find out which events have a high contribution to the medal count.[4]

In our model, Shapley Values measure the overall contribution of each Olympic event to the total medal counts of participating countries since 1992.

A. Analysis and Results

Step 1 Identify the independent and dependent variables

Step 2 Quantified feature contribution

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (4)$$

Where S represents the subset of features, N is the set of all features, and f(S) denotes the model prediction based only on the subset S.

1.By calculating the marginal contribution of each feature to the model output, SHAP values intuitively reflect the influence of the category of items on the number of medals.

2.Truncate the negative SHAP value to 0, ensuring that the contribution value is non-negative.

3.Normalize the contribution value

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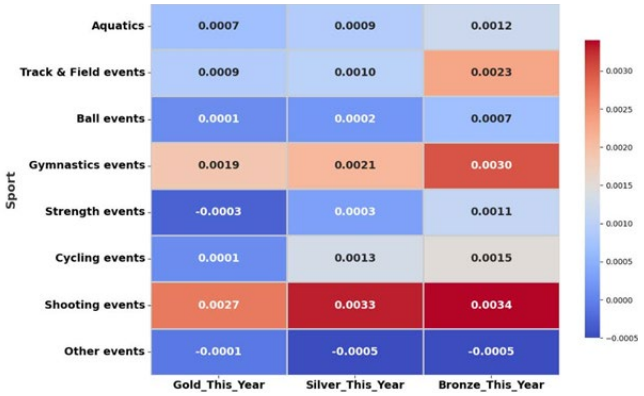


Fig. 5. Shapley Values for Different Sports by Country

Figure 5 visually represents the Shapley values for different sports by country, quantifying how each sport contributes to Olympic medal counts based on the Event Attribution Model.

TABLE V. MODEL RESULTS

continent	*country*	*Advantage Event*	*Shapley Value*
Asia	China	Swimming	0.374
	Japan	Swimming	0.393
	Korea	Archery	0.155
	India	Shootng	0.349
Europe	Russia	Artistic Gymnastics	0.002
	Germany	Shooting	0.087
	France	Fencing	0.211
	Itanly	Fencing	0.294
Africa	Kenya	Athletics	1.000
	Egypt	Boxing	0.192
	Morocco	Football	0.554
Americas	United States	Swimming	0.482
	Brazil	Football	0.012
	Cuba	Boxing	0.132
Oceania	Australia	Swimming	0.705
	New Zealand	Rowing	0.475

In Table V, there are some sports that contribute significantly to the total medal count for most countries: Swimming - Athletics - Weightlifting - Shooting - Boxing.

VI. Great Coach Effect Model Based Based on Bayesian Causal Forest

In Olympic competitions, the quality of a coach can significantly affect a country's medal perfor- mance in certain events. The Great Coach effect refers to the potential increase in the medal tally of a new national team by a great coach. To quantify this impact, we designed a Causal inference model based on a Bayesian Causal Forest

A. Model Building

Based on this, and covariate data is extracted. Bayesian Causal Forests, a recently proposed nonpara- metric causal regression model that employs Bayesian Additive Regression Trees and is specifically designed to estimate heterogeneous treatment effects using observational data. [5]

$$Y_i = \tau(X_i) \cdot T_i + f(X_i) + \epsilon_i \quad (5)$$

•$\tau(Xi)$(Thecausaleffectfunction:)representsthemarginalcontribution ofthegreatcoach to the medal count

• $f(Xi)$ (Function that controls the covariate:) represents the non-treatment effect of the covariate on the medal count.

• ϵi (The error term:) contains factors that cannot be observed.

The Bayesian Causal Forest Algorithm is used to obtain the $\tau(Xi)$ of each input value, that is, to estimate the marginal contribution of the great coach to the number of medals before and after coaching for each country and its sport. At the same time, the marginal contribution effect of the same great coach is averaged to obtain the Average Handling Level (ATE).

B. The Great Coach Effect Level

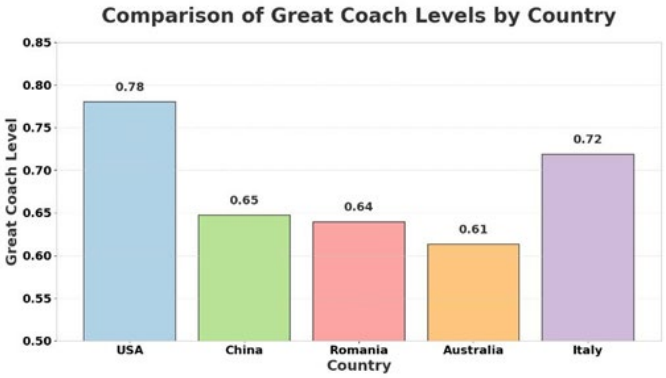


Fig. 6. Dstribution of Medal Counts

The great coach effect levels of the five countries obtained by the model are shown in the Figure 6. It can be found that the great coach effect has different influence degrees in different countries, but in general, it shows significant positive effects. As can be seen from the figure , the United States has the highest level of "great coaches", reaching 0.78, which may be related to the excellent performance of the United States in several sports, indicating that great coaches play a key role in the success of American sports. China followed with a level of 0.74. Romania's low overall level of great coaches shows that even in countries with relatively limited coaching resources, great coaches can have a profound impact[6].

C. Treatment Effect Analysis and Attributionl

The dashed line in the Figure 7 represents the ATE value (0.6729), and the arrow indicates the average causal effect of the treatment effect on the result. The color bar indicates the size of the "Great Coach Effect," and the darker the color, the greater the great coach effect.

The results of ATE analysis can be found in Figure 8.

– ATE vs.X_1:

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In countries with higher levels of sport, the presence of great coaches appears to increase medals more significantly, suggesting that investment in sports infrastructure and human resources may enhance the positive impact of great coaches on medals.

– ATE vs.X_2:

The effect of great coaches is more pronounced in countries with a high number of historical medals, which may mean that countries with a stronger athletic tradition are more likely to benefit from the coaching of great coaches, thereby further boosting their medal count.

– ATE vs.X_3:

The distribution of points in the graph shows the relationship between the great coaching effect and the frequency of historical participation. The positive effect of great coaches is more pronounced in countries with more frequent participation, which may reflect that countries with more experience in specific sports are better able to utilize the expertise and experience of great coaches, thereby increasing their medal chances.

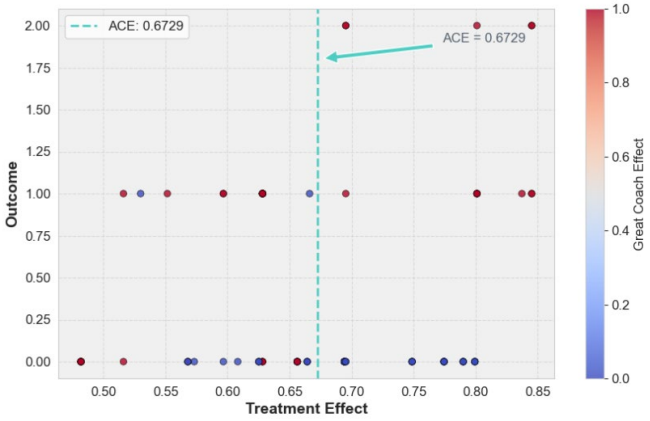


Fig. 7. ACE Analysis

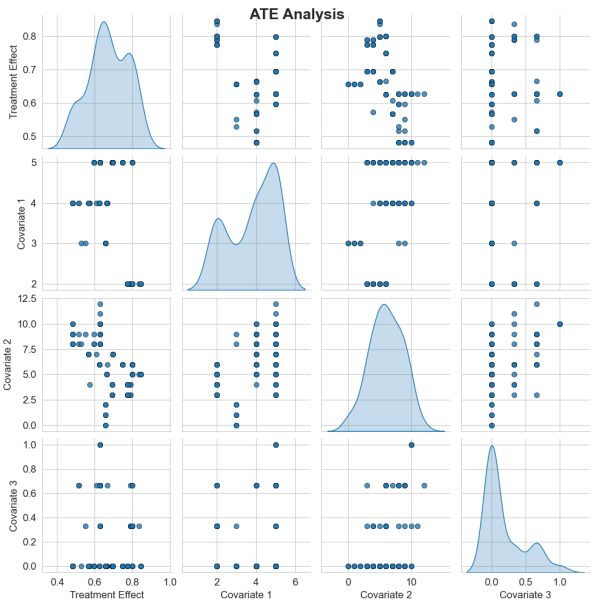


Fig. 8. Treatment Effect vs. Covariates

VII. Conclusion and Suggestion

This study has established a comprehensive modeling framework to address the multifaceted challenge of Olympic medal prediction and strategic analysis. Through the development and integration of four specialized models—BAS-BP Neural Network, Logistic Regression, BAS-BP-SHAP Event Attribution, and Bayesian Causal Forest—we have successfully quantified key determinants of Olympic success and provided actionable insights for national Olympic committees.

A. Conclusion

Our primary contributions are fourfold. Firstly, the BAS-BP Neural Network model demonstrated exceptional predictive accuracy (R2>0.95) for medal counts by effectively synthesizing historical performance data, host nation effects, and nationally clustered sports proficiency levels. The model forecasts a dominant performance by the United States in the 2028 Los Angeles Olympics, projecting 50 gold and 137 total medals, while indicating a potential decline for Australia.[7]

Secondly, the Logistic Regression-based Medal Breakthrough Prediction Model estimated the probability of historically non-medaling nations winning their first Olympic medal. Results indicated generally low probabilities, with a bootstrap-estimated confidence interval of [1, 3] countries likely to achieve this breakthrough in 2028 at a 95% confidence level.

Thirdly, the SHAP value-based Event Attribution analysis revealed that the type of events a nation participates in is a far more significant predictor of medal success than the sheer number of events. Disciplines such as swimming, athletics, and gymnastics were identified as high-contributors. Furthermore, the analysis quantified the strategic advantage host nations gain by introducing new events aligned with their strengths, as observed in France's 2024 and the anticipated U.S. 2028 games.

Lastly, the Bayesian Causal Forest model provided robust, causal evidence for the positive impact of elite coaches—the "Great Coach Effect"—on medal outcomes, calculating an Average Treatment Effect (ATE) of 0.67. This effect was most pronounced in technically complex sports and within nations possessing stronger existing sports infrastructures.

B. Limitation

The BP Model itself has high complexity, poor interpretability in the face of large-scale data, and although it has been optimized by BAS, the BP Model still consumes a lot of computational resources.[8]

The parameters of the trained BP Model are somewhat random, and the output of the model is not always a unique value.[9]

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